

ECO387C - Fall 2024

Problem Set 1

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You may work in groups, but please acknowledge any contributions from others and turn in separate assignments.

Question 1

Recall the exchange economy we considered in class:

Time is discrete and begins at $t = 0$ and goes until period T (assumed to be an **odd** number). There are two types of agents, A and B . Each type has a deterministic endowment pattern. Type A gets one unit of a perishable good in every even period while type B gets one unit of the good in the odd periods. In other words,

$$y^A : 1, 0, 1, 0, \dots$$

$$y^B : 0, 1, 0, 1, \dots$$

Preferences are given by:

$$\sum_{t=0}^T \beta^t \ln(c_t^\ell)$$

for $\ell = A, B$. Each agent faces a real budget constraint of the form:

$$c_t^\ell + b_t^\ell = y_t^\ell + b_{t-1}^\ell(1 + r_{t-1})$$

- a **(PE)** Set up the agent's optimization problem.
- b Derive both agents' Euler equations.
- c **(GE)** Assume that the number of each type of agent is the same, i.e. $N^A = N^B = N$. What is the definition of competitive equilibrium in this economy?
- d Show that in this economy where $N^A = N^B$, the equilibrium interest rate (the real rate of return, $1 + r_t$) is $\frac{1}{\beta}$ in each period.
- e Derive and simplify the intertemporal budget constraint.
- f Solve for the competitive equilibrium allocations $\{c_t^A, c_t^B\}_{t=0}^T$.
- g Now, suppose that $N^A > N^B$. Solve for the competitive equilibrium allocations $\{c_t^A, c_t^B\}_{t=0}^T$.
(**Hint:** Your solution should take the form $c_{\text{even}}^A, c_{\text{odd}}^A, c_{\text{even}}^B, c_{\text{odd}}^B$.)

Question 2

This question is intended to get you started with MATLAB.

- a Use MATLAB to write an OLS subroutine which takes data (inputs) Y and X and returns the OLS estimator $\hat{\beta} = (X'X)^{-1}X'Y$, the variance-covariance matrix $V(\hat{\beta}) = \hat{\sigma}^2(X'X)^{-1}$ and the standard error of the OLS estimates. **Hints:**

- You will need to write your own function. MATLAB's syntax for writing functions can be found [here](#).
- You may find the `inv` command useful.

- b Write a MATLAB subroutine to simulate an AR(1) process for T periods. Specifically, consider the family of AR(1) processes:

$$y_t = \alpha + \rho y_{t-1} + \varepsilon_t$$

where $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$. Note that the unconditional expectation of y_t is $\bar{y} = \frac{\alpha}{1-\rho}$. **Hints:**

- The subroutine's inputs should be α, ρ, σ^2 and T , and it should output a vector $Y = [y_1, y_2, \dots, y_T]$.
 - You may find the MATLAB function `randn` or `normrnd` useful.
 - MATLAB's syntax for writing for-loops can be found [here](#).
 - To append information to a matrix or vector, you can write $Y = [Y; y] : \dots$. This takes the vector or matrix Y and adds the row y .
- c Write a program to analyze the OLS estimate of ρ . You will need the OLS and AR(1) subroutines you wrote in parts (a) and (b).
- Set $\rho = 0.95$, $\alpha = 0$, and $T = 1,100$. Call your AR(1) subroutine.
 - Discard the first 1,000 observations so you're left with 100 left.
 - Use the OLS subroutine to estimate $\hat{\rho}$ by regressing y_t on y_{t-1} .
 - Repeat steps 2 - 3 100 times. Save $\hat{\rho}$ each time. Provide:
 - The mean of the $\hat{\rho}$ s.
 - A histogram of $\hat{\rho}$.
 - Bonus question:** Is the OLS estimate biased in small samples? What if $\rho = 0.5$? What if $\rho = 0.999$?¹

¹You should re-attempt this question when you learn the asymptotic properties of the OLS estimator in ECO388C.